Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

Anonymous CVPR submission

Paper ID *****

Abstract

Replaying a few hundred stored documents comes close to joint training on average when a continual key-information extractor meets new domains, yet fails when its label space grows. We ask why, as an investigation rather than a method. On a class-incremental stream of receipts we test a ladder of hypotheses, each with its prediction stated in advance, and measure where the errors go. Forgetting is not simply harder: with a whole-document memory and a shared softmax the model barely learns the last sessions it is trained on, which naive training learns well. Stale stored labels suffice to explain nearly all of it: giving every stored document its true labels before each session, an oracle no continual learner has, brings the same shared softmax to within 2.4 *points of joint training on every seed (*92.1 *against* 92.0 *full-label F1), although the current documents still call old entities "O". Without true labels, an unbiased loss in the style of MiB recovers about half of the full-label gap, while stopping the sharing of the background class closes most of it, as existing per-session readouts do: with the same memory, one background logit per session reaches* 89.1 *full-label F1 (*88.4 *with raw documents in its place). The principle, familiar from training on partially annotated data, is that an "O" label is evidence only against the classes its annotator could see. Where label growth supplies the routing key, on a single-schema stream, per-session low-rank trunk updates can replace the memory; across schemas they did not, in one run.*

1. Introduction

Key-information extraction (KIE) labels each OCR token of a visually-rich document (form, invoice, receipt), using its words, its 2-D layout and the page image; multimodal encoders such as LayoutLMv3 [13], LiLT [37] and BROS [11] define the state of the art. Deployed extractors meet new document types and growing schemas, while retraining on all history is blocked by cost and by privacy constraints: a continual-learning (CL) problem, whose central obstacle is catastrophic forgetting [24].

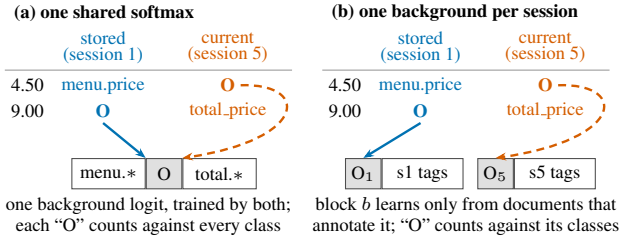


Figure 1. **The background conflict on one pair of price tokens.** A receipt stored in session 1 labels its total "O"; a current receipt of session 5 labels its item price "O". (a) With one shared softmax both "O" labels train the same background logit, against every class. (b) With one background per session (blocks 1 and 5 shown, each with its session's tags), an "O" is evidence only against the classes its annotator could see.

Replay is empirically the strongest remedy in the CL toolbox [23, 38], and on documents it looks sufficient. On LayoutLMv3, a stream of forms and receipts under one fixed label space is learned by plain replay (ER) with 200 stored documents to 87.9 AA, within 3.5 points of joint training on every seed (mean 88.7; Appendix E). Let the label space grow instead, by adding CORD's 30 receipt fields six at a time, and the same method (200 documents) reaches 24.3 AA, against 91.3; both ER cells are legacy-protocol runs (†) that early-stop on the test split, which favours ER (Sec. 3). Distillation, prompts, low-rank adapters and the continual-NER methods built for the shift of the "O" label do no better (Tab. 3). A class-incremental stream is expected to be harder; what is not obvious is why a memory that holds the old documents does not help. This paper is about that question: *why does replay, which solves domain shift, fail when the label space grows?*

We treat it as an investigation (Tab. 1): each explanation a reader would offer gets a prediction stated before its test and an intervention on the same encoder, stream and recipe, and we measure where the final model's errors go (Fig. 2). For-

getting is not just harder (H0): with a whole-document memory that retains under domain shift, the model suppresses the last sessions it is trained on, which naive training does not do. Stale labels in the memory (H1) suffice to explain nearly all of the failure: relabelling every stored document before each session, an oracle, brings the shared softmax to within 2.4 points of joint training on every seed, although the current documents still call old entities "O" (H2). Without true labels, a MiB-style unbiased loss recovers about half of the full-label gap, and removing the shared background (H3, Fig. 1) closes most of it, given a memory; then per-session isolation can replace the memory (H4), but only where label growth supplies a routing key.

None of the ingredients is new. The background shift is known from class-incremental segmentation [4, 6, 7], incremental detection [21, 27] and continual NER [22, 28, 43], and separated or per-session softmaxes [1, 14, 20] already undo the shared normalisation. What has not been done is to take apart, on token-labelled documents, why replay fails: which part a memory repairs, which part survives a memory with true labels, and what the errors of each look like.

Contributions.

1. **The investigation.** A decomposition of why replay fails under label growth, not run before on token-labelled documents: a memory × readout factorial, relabelled memories that isolate the stored-label side (the never-stale one, an oracle, brings a shared softmax to within 2.4 points of joint training), and where the errors go under each readout (Sec. 4).
2. **An empirical finding, not a new mechanism.** A per-session background readout, as in ICE-O and MORE, combined with whole-document replay and a supervision rule in the spirit of training on disjoint tag sets, closes most of the gap to joint training in continual KIE on every seed, where the continual-NER and replay methods in our comparison do not; it works with a raw or a latent memory, and the latent one does as well as raw documents (neither ahead on every seed) while keeping no pixels or token ids and training under half the parameters (Secs. 5 and 5.4).
3. **Boundary conditions.** With the conflict removed, per-session low-rank trunk deltas (SBR-Recall) replace the memory on a single-schema stream, within 2.1 points of joint training on every seed under both metrics, and fail without a memory across annotation schemas (one seed), where the task-identity view says they must (Sec. 5.3); on images, which have no background class, the latent memory is competitive with raw replay while SLCA and RanPAC lead (Appendix F).

2. Related Work

Background shift. When each step annotates only its own classes, the background label changes meaning. In class-incremental segmentation MiB [4] lets a background label mean background or any old class in the loss, and the old model's background mean background or any new class in the distillation; PLOP [7] and SSUL [6] pseudo-label the background with the old model. Incremental detection has named both its stored side [21] and its two-sided form [27]. In continual NER the shift of "O" is the unlabelled-entity problem [22], answered by distillation on "O" tokens [28] and pseudo-labels [43]. We do not claim the conflict; we separate its stored and current sides with relabelled memories and an unbiased loss and measure what each leaves.

Separated readouts and task identity. A softmax over old and new classes together biases them, and separating it removes the bias [1]; ER-ACE [3] masks the old classes out of the softmax of incoming samples. A "none of mine" output per task turns class-incremental learning into within-task prediction plus task identification [15]: MORE [14] isolates each task's parameters and trains each head's out-of-distribution output from saved samples; ICE-O [20], on a frozen encoder with no memory, trains one softmax per session over a constant Other logit. SBR is this pattern with a learned background per session and old blocks kept trainable from a memory; SBR-Recall shares MORE's isolation, by low-rank trunk deltas, with no saved samples.

Continual document IE. Document-level continual learning is recent and small: multimodal LLMs forget under fine-tuning [42], GDI-Bench [18] freezes most parameters, continual table detection [26] uses plain ER, ProtoNER [17] uses prototypes and AMD-Proj [33] projects gradients against task subspaces; Sassioui et al. [34] benchmark layout-aware models by stream regime and find that no method they test consistently preserves old knowledge under label-space shift. They decompose forgetting by regime, we by cause. Our memory is a form of latent replay [9, 29] that keeps whole documents; the broader CL families are surveyed by Wang et al. [38].

3. Setup

Streams and models. A document is a triple of OCR tokens, normalised boxes and the page image; the model emits BIO tags per token, scored by entity-level F1 (`seqeval`). **CIL-CORD**, the stream of every rung: CORD's 30 receipt classes arrive six at a time over five sessions; the head grows, and every entity outside the session, earlier or later, is labelled "O". **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema. **Mixed**: six sessions over three datasets

Table 1. **The hypothesis ladder** (CIL-CORD, LayoutLMv3 unless stated; mean ± s.d. over three seeds, per-seed values when fewer; $^{n=k}$: k seeds so far; †legacy protocol; "–": no full-label run). Verdict: what the row says about its rung's prediction, on every seed; "open": only preliminary, cross-protocol evidence so far. ^{r}True labels for the documents stored before the session just finished, at each boundary, or for every stored document before each session (Sec. 4.2). ^{e}ER also trains every layer and keeps 200 raw documents, so it changes more than the memory format (Sec. 4.1). The pre-session row of H1 also tests H2: it leaves every current document stale.

Rung	Hypothesis	Prediction if true	Intervention (CIL-CORD unless stated)	Full-label F1	AA	Verdict
			Naive fine-tuning (reference)	13.0±0.2	18.3±0.5	
H0	Only drift: forgetting is just harder	a memory that retains under domain shift retains here; raw beats compressed	whole-document memory (CoLaR), shared softmax	30.7±1.5	33.5±1.0	refutes
			raw-document memory (ER), shared softmax^{e}	–	24.3±2.4†	refutes
			raw-document memory, blocked readout	88.4±1.0	88.4±1.1	refutes
H1	Stale labels in the memory	true labels in the stored documents close the gap	older stored documents relabelled at each boundary^{r}	76.5±2.3	63.6±1.9	partly
			every stored document relabelled before each session^{r}	92.1±1.3	91.3±1.3	supports
H2	Current documents mislabel old entities	a loss that lets "O" mean "O or an old class" closes the gap	replay and current losses sliced to their classes	–	60.5±10.1†	open
			MiB-style unbiased CE, both directions	60.0±1.2	72.1±1.9	partly
H3	The shared background normalisation	one background per session closes the gap without true labels, given a memory	CoLaR memory, session-blocked readout (CoLaR-SBR)	89.1±2.0	88.4±2.6	supports
			session-blocked readout, no memory	11.4±2.3	21.6±2.2	supports
			CoLaR memory, blocked readout with a constant background logit	82.4±6.9	80.2±8.1	partly
H4	Isolation replaces memory where a routing key exists	per-session deltas need no memory on one schema, and fail across schemas	per-session deltas, blocked readout, no memory (SBR-Recall)	92.6±1.0	91.9±0.6	supports
			ICE-O (our port): frozen encoder, head per session, no memory	57.0±0.7	35.2±0.7	partly
			the same on Mixed (AA), no memory	–	44.7$^{\dagger,n=1}$	supports
			the same on Mixed (AA), 50-document memory	–	85.9$^{\dagger,n=1}$	supports
			Joint training (reference)	92.0±0.4	91.3±0.8	

and annotation schemas. LayoutLMv3-base [13] is the encoder of every rung; LiLT [37], BROS [11] and text-only BERT appear on DIL. Heads are linear. Table 2 summarises the streams; Appendix A gives their sessions.

Memory and comparison set. Unless stated, a memory is CoLaR (Sec. 5.2): 50 documents per session stored as rank-128 SVDs of their layer-4 activations. The comparison set holds naive fine-tuning and joint training as references and eight methods from five families (Tab. 3): ER [32], DER++ [2], LwF [19], CODA-Prompt [35], CL-LoRA [10], LCA [36], and the continual-NER methods CPFD [43] and IS3 [30]; Appendix C gives our ports and protocols.

Protocol. AdamW, learning rate 5×10^{-5} (LCA keeps its own SGD recipe; Appendix C), batch size 2, seeds 42, 7 and 123. Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100) and

Table 2. **Benchmark settings** (sessions: Appendix A). Train/test documents: FUNSD 149/50, SROIE 626/347, CORD 800/100. AA: mean F1 of the final model over the sessions; full-label F1: on the unmasked CORD test set (Sec. 3).

	CIL-CORD	DIL	Mixed
Datasets	CORD	FUNSD, SROIE, CORD in turn	FUNSD, SROIE, CORD
Sessions	5	3	6
Label space	grows, 6 per session	fixed, unified	grows, 3 schemas
Classes (BIO tags)	30 (61)	4 (9)	12 (25)
At a boundary	new classes	new domain	new classes, domain or schema
Reported	full-label F1, AA	AA	AA

early-stops on 10% held out from its *training* data; the test split never selects a model. Cells marked † come from earlier runs (*legacy protocol*) that early-stop on the test split and vary in epoch cap and precision. A cell is clean only when all three clean seeds exist (Appendix A).

Scoring. The headline number is the *full-label F1*: the final model on the complete CORD test set with all 30 classes, as in continual NER [22]. Only clean runs record it; against a cell with only AA (†, DIL, Mixed) we compare in AA. Retention needs per-session test splits, whose labels show only that session's classes; from $R[i, j]$, the F1 on session j after session i, AA $= \frac{1}{T}\sum_j R[T, j]$; raw scoring of such labels charges a model for every correct entity of another session, so we score each session *session-restricted*, mapping a predicted tag outside session j's label set to "O" before span matching. On every seed both metrics order the main groups of clean runs the same way (Appendix A). Tuning and evaluating on the same scenario overstates CL methods [5]: the comparison set was fixed after a first grid on all three streams, the memory settings were chosen on DIL, SBR's claim threshold (0.5) was not tuned, and the one setting chosen on CIL-CORD is SBR-Recall's delta capacity (Appendices C and D).

4. The investigation

Every rung below runs on CIL-CORD with LayoutLMv3 under the clean protocol and changes one thing against the run it is compared with, unless stated (H0's memory rows change several). Table 1 lists the rungs and Fig. 2 draws them; a verdict word is one the numbers justify on every seed.

4.1. H0: is it just harder forgetting?

Prediction. If label growth only made drift harder, the memory that retains under domain shift would still retain here, a raw memory would do better, and the losses would look like naive forgetting, where old sessions are lost and the newest is learned well.

Test and result: refuted. The CoLaR memory comes within 2.2 AA points of joint training on DIL on every seed (means 87.8† against 88.7†). On CIL-CORD with a shared softmax it reaches 30.7±1.5 full-label F1 (33.5±1.0 AA) against 92.0±0.4 (91.3±0.8). Raw documents do no better (ER, 24.3 AA†, below every CoLaR seed), but ER also trains every layer, keeps 200 documents and is a legacy run; the matched raw-memory comparison is in Sec. 5.2. What separates the memory run from naive forgetting is the session being trained: its newest session ends at 2.0 to 16.0 F1, whereas naive training learns every session to at least 78.6 on every seed. The memory run also freezes layers 1..4 after the first session and replays four stored documents per step; with the same freezing (and, with the memory, the same replay), the blocked readouts with and without the memory learn every session to at least 81.1 (per-seed minimum), so lost plasticity does not explain the suppression. It also keeps 18.7 to 22.6% of the old sessions' spans where naive training keeps none, at an old-span precision of at least 71.6% (a lower bound; Appendix A).

What it rules out, and what it leaves. Drift alone does not explain the failure: the memory that answers drift on DIL leaves the session being trained suppressed here, which naive training, with all of its drift, does not show. Drift is still real: naive training keeps nothing of the old sessions, on a fixed label space every encoder forgets its earlier domains and replay, not EWC or LwF, repairs it; and under naive fine-tuning the drift is largest near the output on all four encoders (Appendix E).

4.2. H1: are the stored labels stale?

Prediction. A document stored at session t carries "O" on every entity of another session, earlier or later, so the memory itself calls background what the model must keep or learn. In detection the same staleness of replayed examples was named foreground shift and blamed for replay's failure [21]. If these stale labels are the cause, giving the stored documents their true labels should close the gap with the shared softmax unchanged.

Test. We refresh the stored labels at two points. *At each boundary*, every document stored *before* the session just finished receives the true labels of all classes seen so far; that session's own documents keep their labels until the next boundary, and no such refresh covers the session trained next. *Before each session*, every stored document receives the true labels of all classes in the head as grown for that session, so no stored label is stale while a session trains; the current documents still call old entities "O". Both read labels no continual learner has and are diagnostics, not methods.

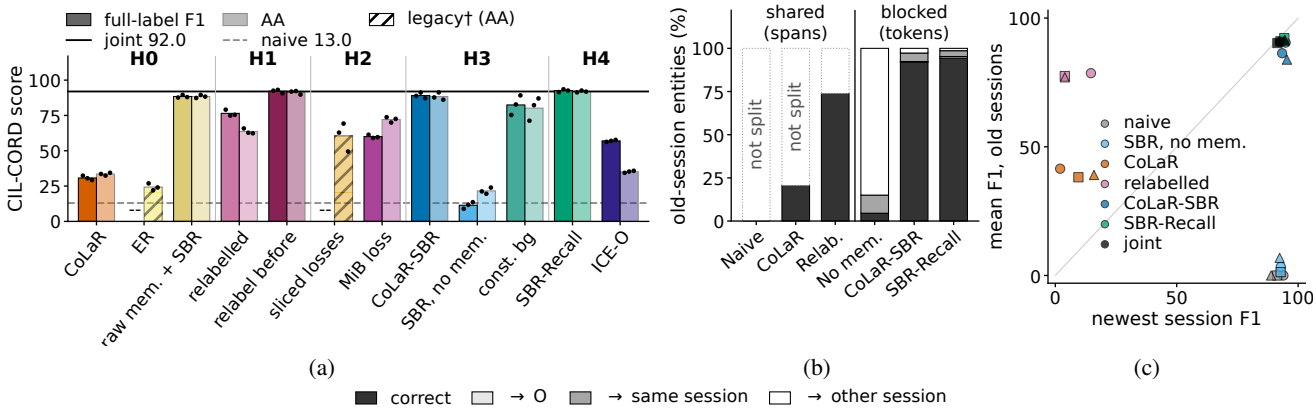


Figure 2. **The findings in one figure** (CIL-CORD, LayoutLMv3, clean protocol unless marked). (a) The ladder (Tab. 1): full-label F1 (dark) and AA (light); bar: mean of three seeds, dots: seeds; hatched: legacy (†), AA only; "–": no full-label run; lines: joint and naive full-label F1. (b) Where the final model puts old-session gold entities (sessions 1 to 4, mean over seeds); a shared softmax records only span recall. (c) Per seed (circle: 42, square: 7, triangle: 123), newest-session F1 against mean old-session F1: with a shared softmax and a memory the newest session is suppressed; naive training learns it and keeps nothing older. More pictures: Appendix D.

Result: partly at the boundary, supported before each session. At the boundary, against the same memory without the refresh, it lifts full-label F1 from 30.7 to 76.5 and AA from 33.5 to 63.6 (means, three seeds): 72 to 79% and 48 to 56% of the seed-matched gap to joint training (92.0, 91.3), and still below CoLaR-SBR (89.1, 88.4) under both metrics. What it cannot reach is the session being trained, whose classes are "O" in every stored document: sessions 3 to 5 are learned only to 2 to 37 F1 on their own diagonal, sessions 3 and 4 rise to 74 to 88 one session later, and the last ends at 3.9 to 14.7, a fifth of AA but 8% of the test spans. Refreshed before each session, the same memory with the same shared softmax reaches 92.1±1.3 full-label F1 and 91.3±1.3 AA, within 2.4 points of joint training on every seed under both metrics and above CoLaR-SBR on every seed (Tab. 1).

What it rules out. Stale stored labels suffice to explain nearly all of the failure of replay under a shared softmax, on the session being trained as much as on old ones (three seeds). The control is an oracle: it shows what the memory must hold, not how a continual learner could obtain it.

4.3. H2: do current documents mislabel old entities?

Prediction. A current document labels every entity of an earlier session "O", so even a memory with true labels is opposed in every batch by documents that call the same strings background. This is the second side of the conflict named in detection [27]. If this pull is the cause, a loss that lets a current "O" mean "O or an earlier class", as MiB does for segmentation [4] and marginal-likelihood taggers do offline [8], should close the gap within a shared softmax.

Test. Two clean controls test it: the never-stale memory of H1, which leaves every current document stale, and, with the memory, the shared softmax and no true labels, MiB's unbiased cross-entropy in both directions: a current "O" token matches "O" or any earlier-session class, a replayed one "O" or any class introduced after storage (no distillation).

Result: not needed, and a loss repair only partly helps. With true stored labels the gap closes although every current document still calls old entities "O" (H1), so the current side is not needed to explain the failure. The MiB-style loss lifts full-label F1 to 60.0±1.2 and AA to 72.1±1.9, above CoLaR and below joint training on every seed under both metrics: 46*to*49% of the seed-matched gap in full-label F1 and 64*to*70% in AA, and below CoLaR-SBR on every seed. It learns every session to at least 90.1 on its diagonal, so it removes the suppression of the session being trained, but the first session ends at no more than 35.5 F1 on any seed.

4.4. H3: is it the shared normalisation?

Prediction. If the conflict lives in the shared normalisation, one background logit trained by documents that disagree about what is background, then one background logit per session, normalised with that session's tags only, should close most of the gap with no true labels. This is SS-IL's diagnosis of a shared softmax [1] with a background added, and ICE-O's readout [20] without its frozen encoder. It needs the memory: separating the labels does not undo drift, so without a memory the same readout should fail, and fail the way drift does.

Result: supported, on every seed. With the memory the session-blocked readout (CoLaR-SBR, Sec. 5.1) reaches

89.1±2.0 full-label F1 and 88.4±2.6 AA, 0.6 to 5.3 full-label points below joint training per seed, above the memory relabelled at each boundary and below the memory relabelled before each session on every seed, under both metrics. The owning block claims 95.4 to 98.3% of the old entity tokens, and most remaining errors (56.6 to 71.6% of the wrong tokens) give the token another tag of its own session. Without the memory the same readout scores 11.4±2.3 (21.6±2.2 AA), and its errors go elsewhere: not one old entity token is predicted "O", 83.8 to 93.5% of the wrong ones receive another session's class, and each old block claims 17.0 to 24.3% of the tokens of its own split that are not its entities, against 0.6 to 1.7% with the memory. Old blocks on a drifted trunk claim far more than their own entities; that is drift, and the memory reduces it.

What goes against it. It is neither the only repair nor the best one: true stored labels bring a shared softmax closer to joint training, above CoLaR-SBR on every seed; the blocked readout needs no true label. The readout does nothing alone, and it is not free: on DIL, where there is nothing to separate, CoLaR-SBR is below CoLaR (AA $86.2^{\dagger}$ against $87.8^{\dagger}$).

What discriminates what. Drift (H0) is told apart from the label hypotheses by the session being trained, suppressed under a shared softmax with a memory and learned without one. Within the label hypotheses the never-stale memory decides: the stored side (H1) suffices to explain nearly all of the failure, and the current side (H2) is not needed. We read the shared normalisation (H3) as the channel through which stale labels act, as negatives for every class; then separating it should work even with a background that does not learn. With every background logit fixed at zero, as in ICE-O, the memory and blocked readout stay above CoLaR and below joint training on every seed under both metrics, recover most but not all of CoLaR-SBR's gain over CoLaR on seeds 42 and 7, and are above CoLaR-SBR on seed 123 (Tab. 1): the reading is partly supported; separation does most of the work, and the learned background adds to it on two of three seeds.

5. From diagnosis to design

The principle. The rungs point at one fact about annotation under label growth, familiar from training on partially annotated data [8]. A token marked "O" in session s means "none of the classes annotated in session s": it is evidence against those classes and silent about every other. A shared softmax turns it into evidence against all classes at once (H3), and stale stored labels are the same error made by the memory (H1). The principle: *supervise each class only with negatives from documents that annotated it, and normalise together only classes that were annotated together.*

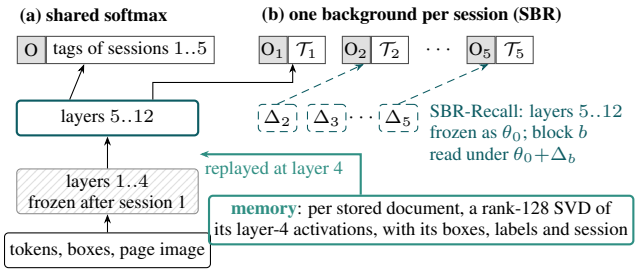


Figure 3. **The two readouts on one encoder and memory.** (a) One softmax over all tags and one background. (b) SBR: one background logit per session, trained only by documents that annotate it (Eq. (2)); the largest claim $e_b = 1 - p_b(\mathrm{O}_b) > 0.5$ labels the token. SBR-Recall drops the memory and reads block b under its session's delta Δ_b (Eq. (3)).

Three consequences follow: each session needs its own background; the memory must keep whole documents with their own labels and the session that wrote them; and each session's background logit, trained on that session's true negatives, is also a detector of its entities, so it can route a token without task identity.

5.1. SBR: one background per session

SBR is our instantiation of a per-session background readout [14, 20]. Every session that brings new tags opens a block b holding those tags $\mathcal{T}_b$. A shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$. SBR gives every session its own background logit $o_b(x) = w_b^\top h(x) + c_b$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\right), \tag{1}$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block b a token labelled "O" in session-b data is truly background, so the block's supervision is never stale. For a token x with label y in a document of session s,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\ \mathrm{CE}\big(p_b(\cdot \mid x),\, y_b\big),$$
$$y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: $m_{sb}{=}1$ when session s annotates every tag of $\mathcal{T}_b$; $m_{sb}{=}1$ with $y_b{=}\mathrm{O}_b$ when b belongs to a different annotation schema (every token is a negative for b, a training-time convention; Appendix B); and $m_{sb}{=}0$ otherwise, because the document's "O" labels say nothing about block b. Stored documents follow the rule of the session that stored them. A session that adds no tags creates no block, so on a fixed label space SBR has a single block. At inference each block reports its claim $e_b = 1 - p_b(\mathrm{O}_b \mid x)$;

the block with the largest e_b labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise, as in ICE-O's "Other unless some session scores above it". Blocks trained at different times need not produce claims on one scale, so we measure the routing directly (Fig. 2b). Figure 3 draws both readouts; details: Appendix B.

5.2. The memory

SBR's supervision needs stored documents that keep each token's label at its position and the session that wrote it. CoLaR (Compressed Latent Replay) keeps whole documents: after the first session layers $1..k$ are frozen, and each stored document's layer-k activations H_j are kept as a rank-r truncated SVD with its boxes, mask, labels and session, rebuilt at replay and injected at layer k; it holds no raw pixels or token ids (no privacy claim: activations can leak content). On DIL, with 50 documents per session, it retains $87.8\pm0.5^{\dagger}$ AA against $87.9\pm1.4^{\dagger}$ for ER with 200 raw documents. The memory type does not matter for the finding: raw documents under the same readout reach 88.4 ± 1.0 full-label F1 against 89.1 ± 2.0, seed-matched differences (raw minus latent) of -1.7 to +0.9 points (-1.9 to +2.2 in AA). We use CoLaR because it keeps no pixels or token ids and trains fewer layers (Sec. 5.4).

5.3. SBR-Recall: the test of H4

Prediction (H4). With the conflict removed, the memory is needed only against drift, and the task-identity view [15] says when it can go: if each block is read under the trunk state that learned it, nothing drifts for that block, provided the model can tell which state a token belongs to. On one schema under label growth each block's background logit supplies that key. Across schemas the all-background terms of Eq. (2) train every old block towards background on other-schema documents with nothing to pull it back, so isolation should fail without stored documents; on a fixed label space there is one block and nothing to route by.

The intervention. SBR-Recall freezes the layers above k after the first session (the base θ_0), stores for each later session b a rank-64 delta Δ_b [12] of them ($\Delta_1{=}0$), and computes every term of Eq. (2) for block b, at training and test time, under the trunk state that learned it (Appendix B):

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b,\, z_{\mathcal{T}_b}\,]\right)\left(x;\, \theta_0 + \Delta_b\right). \qquad (3)$$

Result: supported, with its predicted failure. On CIL-CORD SBR-Recall reaches 92.6 ± 1.0 full-label F1 (91.9 ± 0.6 AA) with no stored document, within 2.1 points of joint training on every seed under both metrics; its owning block claims 97.0 to 98.3% of the old entity tokens. Our port of ICE-O (frozen encoder, one head per session over a constant Other logit, no memory) forgets nothing (final row within 0.3 points of its diagonal on every seed) and reaches 57.0 ± 0.7 full-label F1 (35.2 ± 0.7 AA), below SBR-Recall on every seed under both metrics; under our 10-epoch cap sessions 4 and 5 were still improving and session 3 early-stopped at 0.0 F1 on every seed, so this is a lower bound for ICE-O. On Mixed without a memory it scores 44.7 $\mathrm{AA}^{\dagger}$ (seed 42): the blocks of the other schemas fall silent until their domain returns, as predicted, and with the 50-document memory restored it scores 85.9 (Appendix D).

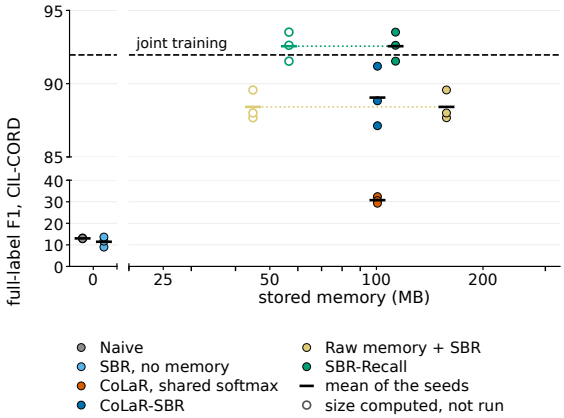


Figure 4. **Accuracy against stored bytes** (CIL-CORD, clean protocol; dots: seeds, bar: their mean). Bytes: what a method keeps across sessions; joint training keeps every document. Filled: the size the run held; hollow: the same run's size computed for an encoding it never ran in (raw documents with 8-bit pixels, SBR-Recall's deltas at two bytes per parameter); a dotted line joins the two sizes of one run. Sizes: Tab. 5.

5.4. What the memory costs

CoLaR stores 402.7 KB per document (100.7 MB for the 250 documents of CIL-CORD; Fig. 4), 2.8× less than the fp16 layer-4 activations it compresses. Raw replay keeps 630.8 KB per document as our code holds it, with fp32 pixels and int64 ids and boxes (157.7 MB), and would take 44.8 MB with 8-bit pixels. SBR-Recall keeps no document but one delta per session: 28.3 MB in fp32 as trained (113.2 MB in all), or 14.2 MB (56.6 MB) at two bytes per parameter, a reporting choice that no run evaluated.

Bytes do not separate the methods; the readout does. Every store read through the blocked readout (latent or raw documents, or deltas) ends within 5.3 full-label points of joint training on every seed, while the latent memory under a shared softmax stays at 30.7 ± 1.5 and the blocked readout alone at 11.4 ± 2.3. At rank 128 CoLaR saves no bytes: it is 1.6× smaller than raw documents as held but 2.2× larger than with 8-bit pixels. What it gives instead is a whole-document memory without pixels or token ids, 55% fewer trainable parameters than raw replay (layers $1..4$ stay frozen and replay runs through the upper eight layers only) and,

Table 3. **Benchmark results on LayoutLMv3**, grouped by family. Memory: what a method keeps (sizes as recorded on CIL-CORD, SBR-Recall's deltas at two bytes per parameter). Readout: one shared softmax or one block per session. Mean ± s.d. over three seeds, per-seed values when fewer; "–": not run. †legacy protocol; $^{n=k}$: k clean seeds so far; ^{c}one fp16 LwF seed that collapsed to all-background is excluded; ^{b}with the 50-document memory; ^{p}made before we fixed our port's orthogonality penalty. Ports and protocol per cell: Appendix C.

				CIL-CORD		Mixed	DIL
Family	**Method**	**Memory (what is stored)**	**Readout**	full-label F1	AA	AA	AA
Reference	Naive fine-tuning	none	shared	13.0±0.2	18.3±0.5	44.4/43.7†	41.3±1.5†
	Joint training	all data, all sessions	shared	92.0±0.4	91.3±0.8	86.9/88.5†	88.7±1.1†
Regularisation	LwF (2016)	previous model	shared	–	18.4/18.5†,c	47.7†,c	43.7±1.1†
Replay	ER (2019)	200 raw documents	shared	–	24.3±2.4†	75.6/73.3†	87.9±1.4†
	DER++ (2020)	200 raw documents + logits	shared	–	20.4±1.4†	75.1/74.0†	88.0±0.9†
Prompt, LoRA	CODA-Prompt (2023)	none (prompt pool)	shared	–	14.0/15.9†	12.1/10.6†	32.5±1.2†
	CL-LoRA (2025)	previous model, past adapter matrices	shared	–	18.7/18.6†	45.7/43.9†	41.1±0.3†,p
Continual NER	CPFD (2023)	previous model	shared	–	2.7/0.0†	0.3/15.1†	6.1±7.3†
	IS3 (2024)	previous model, class prototypes	shared	–	5.6/7.7†	0.0/40.6†	38.3†
Merging	LCA (2026)	base and per-session backbones, class Gaussians	shared	–	13.7/7.0†	38.0/36.4†	43.9±2.5†
This paper	CoLaR	50 docs/session, rank-128 layer-4 latents (100.7 MB)	shared	30.7±1.5	33.5±1.0	75.6/74.6†	87.8±0.5†
	CoLaR-SBR	50 docs/session, rank-128 layer-4 latents (100.7 MB)	blocked	89.1±2.0	88.4±2.6	**89.1±0.6**	86.2±1.3†
	+ LexSlot head slots	+ head slots, token ids	blocked	–	89.7±0.5†	89.6±1.0†	87.0±0.4†
	SBR-Recall	one rank-64 trunk delta per session (56.6 MB)	blocked, per delta	**92.6±1.0**	**91.9±0.6**	85.9†,b	–

under the same readout, the accuracy of raw replay, neither ahead on every seed. The rank is the one knob that sets its size; Appendix B gives the sizes at lower ranks, computed, not evaluated. SBR-Recall's cost grows with the sessions, not with their documents, so it is the cheaper store only when a session's documents would take more than its delta.

What each component adds. On the runs of Tab. 1 (Tab. 7), and on every seed, the memory alone raises full-label F1 over naive training by 16.3 to 19.2 points, and the blocked readout alone moves it by −4.3 to +0.7; together they raise it by 74.1 to 78.4, more than the two alone add up to, so the gain is an interaction, not a sum. The memory's type does not matter for it: raw documents in its place are neither above nor below CoLaR-SBR on every seed (Sec. 5.2). Per-session deltas in place of the memory are above CoLaR-SBR under both metrics. Design-choice ablations: Appendix D.

6. Evidence beyond one stream

The comparison set. On CIL-CORD every method of Tab. 3 with a shared softmax stays at or below CoLaR in AA (33.5). Four baselines end below naive training on every seed because they learn later sessions poorly (Appendix C); all rows are our ports onto one growing token head. On Mixed, where both causes act, CoLaR-SBR reaches 89.1±0.6 AA under the clean protocol, above ER (74.5†) on every seed although the legacy protocol favours ER; joint training has only a legacy cell there (87.7†), so we do not rank the two. Beyond CIL-CORD the evidence is the blocked readout with a memory on Mixed (three clean seeds), drift and its repair by replay on DIL (four encoders, legacy) and SBR-Recall's failure across schemas (one seed); the controls of H1 to H3 and the memory comparison ran on CIL-CORD only.

7. Limitations and conclusion

Limitations. The study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones; the controls of the ladder ran on LayoutLMv3 and one class-incremental partition of CORD only (Sec. 6). SBR-Recall on Mixed, the design-choice ablations and the image probe are single- or few-seed diagnostics. We did not run several comparators from this literature (among them PLOP, BPF, MiB's distillation and per-class sigmoid heads; Sec. 4.3). Joint training is a reference, not a bound. SBR-Recall grows by one delta and one pass per session and freezes its base after session 1, so it may depend on the session order. CoLaR's memory grows with the sessions and at rank 128 takes more bytes than raw documents with 8-bit pixels; storing no raw pixels or token ids is weaker than privacy.

Conclusion. Replay fails under label growth because two things go wrong and a memory repairs only one, drift. The other, the background conflict, lies almost entirely in stale stored labels: an oracle that keeps them true brings a shared softmax to within 2.4 points of joint training on every seed. Without true labels a MiB-style loss recovers about half of the full-label gap, and a per-session readout most of it, by letting an "O" label count only against the classes its annotator could see; where label growth supplies a routing key, per-session deltas then replace the memory. Continual document methods should be tested on a growing label space, with a mask-free metric.

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Appendix

Table 4. **Sessions.** Top: CIL-CORD (classes with the prefix of their group); pool: CORD training receipts that contain one of the session's entities (10% of each pool is held out for early stopping under the clean protocol). Bottom: Mixed; schema: the annotation scheme SBR's supervision rule reads (Appendix B).

CIL-CORD (30 classes, 61 BIO tags)

S.	Classes	Pool
1	menu: cnt, discountprice, itemsubtotal, nm, num, price	800
2	menu: unitprice, vatyn, sub.cnt, sub.nm, sub.price, sub.unitprice	382
3	sub_total: discount_price, etc, othersvc_price, service_price, subtotal_price, tax_price	549
4	total: cashprice, changeprice, creditcardprice, emoneyprice, menuqty_cnt, menutype_cnt	723
5	total.total_etc, total.total_price, void_menu.nm, void_menu.price, sub.nm, sub.cnt	780

Mixed (25 tags)

S.	Dataset: tags introduced	Schema	Pool
1	FUNSD: HEADER, QUESTION	funsd	149
2	FUNSD: ANSWER	funsd	147–148
3	SROIE: COMPANY, DATE, ADDRESS, TOTAL	sroie	625–626
4	CORD (super-classes): menu, sub_total	cord	799–800
5	CORD (super-classes): total, void_menu, sub	cord	797–798
6	FUNSD: none (all FUNSD labels)	funsd	149

A. Setup and protocol

Streams. We use the dataset-provided OCR and boxes of FUNSD, CORD and SROIE, page images at 224×224 and 512 text tokens, so a LayoutLMv3 document is a sequence of 709 positions. On CIL-CORD (Tab. 4) a session's training and test pools keep every receipt that contains at least one of its entities, every entity of another session is labelled "O", and the head grows by twelve BIO tags per session; joint training uses the full-label pool, each receipt once with all its labels. Mixed alternates between new classes within a dataset and a new dataset, and its last session returns to FUNSD with all its labels and adds no tag. DIL keeps one schema {HEADER, KEY, VALUE, OTHER} fixed over FUNSD→SROIE→CORD. Every head is one linear layer; when it grows, old rows are kept and new rows are drawn from $\mathcal{N}(0, 0.02^2)$.

Scoring. A session's test labels show only its own classes, so raw scoring charges a model for every correct entity of another session and penalises it for knowing more classes. Session-restricted scoring maps every predicted tag outside session j's label set to "O" before `seqeval` span matching (for SBR after the block decision of Eq. (4)); a predicted entity can then be a false positive only in its owning session and a gold entity a miss only in its own, so each error is charged at most once. Its gap is coverage: a session's split keeps only receipts that contain one of its entities, so false positives on other receipts go uncounted. Full-label F1 scores the final model on the unmasked CORD test set and has no such gap. On every seed the order no memory (naive training, blocked readout) $<$ CoLaR $<$ memory relabelled at the boundaries $<$ {CoLaR-SBR, SBR-Recall, memory relabelled before each session, joint training} holds under both metrics; the MiB-style loss and ICE-O lie between CoLaR and the top group under both, and only the MiB-style loss and the boundary-relabelled memory swap places between the metrics. For CoLaR-SBR, SBR-Recall and joint training the two metrics differ by at most 1.3 points on every seed; for weak models AA is the more lenient, by 4.8 to 12.3 points without a memory and 1.2 to 5.2 for CoLaR. AA weights the five sessions equally, whereas session 1 holds 57% of the 1,306 gold spans. For the shared softmax we have span counts but no per-token outputs. Given the exact true positives and misses, full-label F1 fixes the number of false-positive spans, and (misses − false positives)/misses would bound the share of missed spans labelled entirely "O" if every wrong span overlapped at most one missed gold span. Long runs of wrong tags break that proviso (the quantity reads up to 81% for the blocked readout without memory, whose per-token outputs put no old entity token at "O"), so we call it an indicator. The same count does bound CoLaR's old-span precision from below, at 71.6% on every seed.

Protocol. The clean protocol trains every session in fp32 at batch size 2 with gradient checkpointing, AdamW (learning rate 5×10^{-5}, weight decay 0.01, clipping 1.0), at most 10 epochs (100 for joint training), and early-stops (patience 2, 0.1 F1 points, best weights restored) on 10% held out from the session's training pool, scored with the same session restriction; the test split never selects a model. Legacy (†) cells early-stop on the test split, differ in epoch cap by method and partly ran in fp16 mixed precision (Appendix C). A cell switches to the clean protocol only when its three clean seeds (42, 7, 123) exist, so no cell mixes the protocols. The clean, gate and rescore runs ran one at a time on one 6 GB RTX 2060; some rows of the first DIL grid ran on a rented GPU.

B. Method details

CoLaR. After the first session the embeddings and layers $1..k$ are frozen, so a stored activation is produced by the same layers forever and cannot go stale. At the end of each session the layer-k output of 50 training documents is kept as the fp16 factors $(U_r\Sigma_r)$ and $V_r^\top$ of its truncated SVD, with the document's boxes, attention mask, labels and storing session; no raw pixels and no token ids are kept. At every step four stored documents are rebuilt, injected at layer $k{+}1$ by a forward pre-hook (their boxes and mask drive the layout biases and attention mask), and their cross-entropy is added with weight 1; gradient checkpointing is off for the replayed pass. A document is counted as $2r(709{+}768)$ bytes of factors plus 24,576 bytes of boxes, mask and labels (and one byte for the session under SBR), which for the 250 documents of CIL-CORD gives the 100.7 MB of Tab. 7 (1 MB

$= 10^6$ bytes). The rank r is the one knob that sets this size: at rank 64 or 32 a document would take 213.6 or 119.1 KB (53.4 or 29.8 MB in all), computed from the same count; we make no accuracy claim for these ranks. The record the implementation holds in memory is about 1% larger (406,800 against 402,689 bytes per document), because it also holds the labels in the final label space; no method reads that field, only the relabelled-memory diagnostics of rung H1.

SBR. A session that introduces at least one tag not yet covered opens a block with those tags, a background logit (a linear map from the 768-dimensional head input to one logit) and its session's schema; a session that only re-annotates known tags opens none, so on DIL SBR has one block. For a document of session s the supervision rule gives each block b one of three modes: *full* when s annotates every tag of b (target y on b's tags, O_b elsewhere); *all-background* when b belongs to another schema (target O_b on every token, a training-time convention that, between related receipt schemas, can call a real entity background); and *skipped* otherwise. On CIL-CORD a document therefore trains only its own block; on Mixed the all-background mode keeps a block from claiming another dataset's tokens. A stored document follows the rule of the session that stored it. At inference, with $e_b = 1 - p_b(\mathrm{O}_b \mid x)$ and τ=0.5 (ties to the lower block index),

$$\hat{y}(x) = \begin{cases} \arg\max_{t \in \mathcal{T}_{b^\star}} p_{b^\star}(t \mid x) & e_{b^\star} > \tau, \\ \mathrm{O} & \text{otherwise}, \end{cases} \qquad b^\star = \arg\max_b e_b. \tag{4}$$

SBR-Recall. Every linear layer named query, key, value or dense (the attention output and both feed-forward layers) in layers k+1..12 carries one low-rank pair per delta, rank 64 with α=128, A Kaiming-uniform and B=0, so a fresh delta is the identity. A rank-64 delta over the eight layers above k=4 has $8 \times [3 \cdot 64 \cdot 1536 + 64 \cdot 1536 + 2 \cdot 64 \cdot (768{+}3072)] = 7{,}077{,}888$ parameters; the four deltas of CIL-CORD, counted at two bytes per parameter, give the 56.6 MB of Tab. 7 (14.2 MB per session), and 113.2 MB (28.3 MB per session) in fp32. The two-byte count is a reporting choice: the deltas were trained and evaluated in fp32, and no run evaluated half-precision deltas. The first session trains the base; afterwards the encoder is frozen and only the head, the background weights and the current delta train. The loss groups a document's blocks by the delta each is read under and runs one forward pass per distinct delta, so every term is computed under its block's own trunk state, as is inference (five passes on CIL-CORD; a cached implementation repeats only the $12 - k$ upper layers, about 3.7 layer-equivalents of one pass). Without a memory on one schema a later session's loss contains only its own block, so an old block's delta and background weights are outside the graph and its rows of the tag head get zero gradient; only AdamW's decoupled weight decay (under 1% over a run) and out-claiming by new blocks can move its tokens. In the clean runs no session falls more than 0.6 points below its diagonal at any later boundary.

Memory accounting (Tab. 5). The raw-replay row counts a document as our code holds it: $3 \cdot 224^2$ pixels at four bytes (one byte in the 8-bit row, which no run used) and $512 \cdot 7{+}1$ int64 values (per token its id, four box coordinates, mask and label; one for the session). The uncompressed latent is $709 \cdot 768$ fp16 values plus the 24,576 bytes of the CoLaR record. Raw replay with the blocked readout trains the whole network; CoLaR and SBR-Recall train only what lies above layer 4 after session 1, and CoLaR replays a document through the upper eight layers only. Wall-clock is reported for completeness: only one CoLaR-SBR run ran uninterrupted, so we do not compare run times.

C. Baselines

Ports. Every baseline is ported onto the one growing token head that every other row uses (Tab. 6); none is checked against its published NER numbers. Hyper-parameters are the methods' defaults as recorded by each run: ER and DER++ keep a reservoir of 200 documents (DER++ α=β=0.5), LwF uses T=2, CL-LoRA a rank-8 adapter with an orthogonality penalty of 0.5. LCA keeps its published optimiser instead of our AdamW recipe: SGD with momentum 0.9, a cosine schedule, learning rate 0.01 and weight decay $5{\times}10^{-4}$, and SGD at $5{\times}10^{-3}$ for the realignment. Rows marked ^{p} in Tab. 3 were made before we fixed the order of the adapter matrices in our ports' orthogonality penalty.

How the set was fixed. The set was fixed after a first grid on all three streams. There EWC [16], L2P [41] and DualPrompt [40] left at least sessions 4 and 5 of CIL-CORD unlearned on every seed, and ER with SAM (our C-Flat++ port, run with its curvature weight at 0) stayed within 2.8 AA of ER on every seed, so these were dropped; O-LoRA [39] was dropped because its runs predate the penalty fix. Methods that fail on the scenario (CPFD, IS3) were kept. Four baselines end below naive training on every seed because they learn later sessions poorly: CPFD learns no session after the first (at most 0.0 F1 on a later diagonal), and IS3, CODA-Prompt and LCA each learn some later session to at most 32.3 F1, where naive training learns every session to at least 78.6.

Table 5. **Memory accounting** (CIL-CORD: 250 stored documents, or four deltas; 1 MB = 10^6 bytes). Computed: the size of an encoding no run used, from the counts in the text. Trainable: parameters trained after session 1; peak GPU memory and hours per run from the clean runs ($^{n=k}$: only k uninterrupted runs timed). Generated.

Storage form	Per unit	Total MB	Pixels	Token ids	Trainable (M)	Peak GPU MB	h/run
Layer-4 latent, fp16, uncompressed (computed)	1113.6 KB/doc	278.4	no	no	–	–	–
Raw document as held (fp32 pixels)	630.8 KB/doc	157.7	yes	yes	125.4	2666	2.9
Raw document, 8-bit pixels (computed)	179.2 KB/doc	44.8	yes	yes	–	–	–
CoLaR rank 128	402.7 KB/doc	100.7	no	no	56.8	2436	$2.0^{n=1}$
SBR-Recall deltas, fp16 (computed)	14.2 MB/session	56.6	no	no	–	–	–
SBR-Recall deltas, fp32, as trained	28.3 MB/session	113.2	no	no	7.1	2436	1.7

Table 6. **Baseline card** (the methods of Tab. 3). What our port changes against the published method, what it keeps across sessions, and the number of seeds behind each cell (†: legacy protocol). Legacy cells early-stop on the session's test split; their epoch cap is 100 for ER, DER++, LwF, CPFD and IS3, 30 for CODA-Prompt and CL-LoRA and 10 for LCA and the CoLaR family; most ran in fp16 mixed precision, but IS3 and LCA always in fp32, and seed 42 of several CIL-CORD baselines in fp32, so some cells mix precisions across seeds. LwF excludes one fp16 seed per scenario that collapsed to all-background. The first DIL grid saved no configuration; its settings are the grid script's defaults. Generated.

				Seeds		
Method	Family	What our port changes	Memory	CIL-CORD	Mixed	DIL
ER	Replay	–	200 raw documents	$3^{\dagger}$	$2^{\dagger}$	$3^{\dagger}$
DER++	Replay	–	200 raw documents + logits	$3^{\dagger}$	$2^{\dagger}$	$3^{\dagger}$
LwF	Regularisation	–	previous model	$2^{\dagger}$	$1^{\dagger}$	$3^{\dagger}$
CODA-Prompt	Prompt, LoRA	prompts prepended to the input sequence (an input-prepend variant)	none (prompt pool)	$2^{\dagger}$	$2^{\dagger}$	$3^{\dagger}$
CL-LoRA	Prompt, LoRA	one shared rank-8 adapter retrained each session (published: random-orthogonal shared and block-wise task-specific adapters, early-exit distillation); gradient reassignment at scale 1, i.e. off	previous model, past adapter matrices	$2^{\dagger}$	$2^{\dagger}$	$3^{\dagger}$
CPFD	Continual NER	one growing token head (published: per-task classifiers); sample weights set to 1	previous model	$2^{\dagger}$	$2^{\dagger}$	$3^{\dagger}$
IS3	Continual NER	one growing token head (published: per-task classifiers)	previous model, class prototypes	$2^{\dagger}$	$2^{\dagger}$	$1^{\dagger}$
LCA	Merging	one growing token head (published: per-task heads); Gaussians per token class, "O" excluded	base and per-session backbones, class Gaussians	$2^{\dagger}$	$2^{\dagger}$	$3^{\dagger}$

D. Extended results and ablations

Retention (Fig. 5). Read the diagonal (each session when it is learned) and the bottom row (the final model). Naive training learns every session and keeps none of the earlier ones. CoLaR keeps part of the old sessions, but from session 3 on its diagonal is low: the session being trained is suppressed (rung H0). With the relabelled memory sessions 3 and 4 rise one boundary after they are learned, while the last session stays low (rung H1). The blocked readout without a memory learns every session and loses the old ones; with the memory the columns hold. *Where the remaining gap lies:* under CoLaR-SBR a column loses 1.5 to 9.0 points from its diagonal to the final row of the mean matrix, session 2 the most, and under SBR-Recall every column of the mean matrix ends within 1.4 points of its diagonal, so what CoLaR-SBR still loses to joint training is retention of early sessions, which per-session deltas remove on this stream.

Per class (Fig. 6). The dump re-runs are not table cells, and every statement here is about seed 42 only. Of the 23 classes that occur in the test set, the joint-training re-run finds 21 (F1 above 0), the CoLaR-SBR re-run 20, the CoLaR re-run 15 and the naive re-run 1; 2 of the classes the CoLaR-SBR re-run misses are missed by joint training too. In the CoLaR re-run the session-5 classes are almost never found (the suppressed newest session) and the session-1 classes, which hold most test entities, are found only in part, while several session-2 and session-3 classes are kept.

Design choices (Fig. 7 and Tab. 8). Figure 7 draws the component grid with its seeds; Tab. 8 gathers the design-choice runs, all legacy gate runs and mostly single seeds, so each is compared only within its own protocol. *Delta capacity.* At rank 8 on query, key and value, SBR-Recall learns its weakest session (the sub-total and total fields, mostly prices) only to the value in the min $R[j, j]$ column, and adding the memory does not help; at rank 64 with the attention output and feed-forward layers every session is learned on all three seeds. Isolation was sound at rank 8; the capacity of the delta was the limit. Rank 64 is the one setting we chose on CIL-CORD, from these legacy runs, before the clean re-runs. *Across schemas.* Without a memory on Mixed, an old block is trained towards its background on every token of another schema's documents and nothing trains it towards its own tags, so its blocks fall silent until their domain returns and at

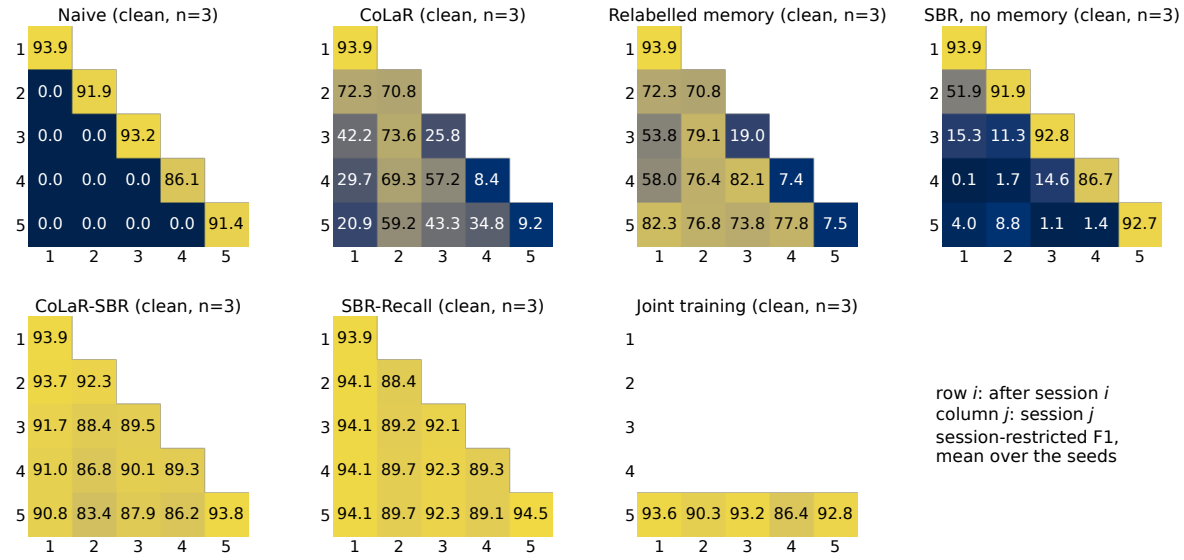


Figure 5. **Retention matrices** (CIL-CORD, LayoutLMv3, clean protocol, session-restricted F1, mean over the three seeds). Row i: after training session i; column j: session j. Joint training is one pooled run, so only its final row exists.

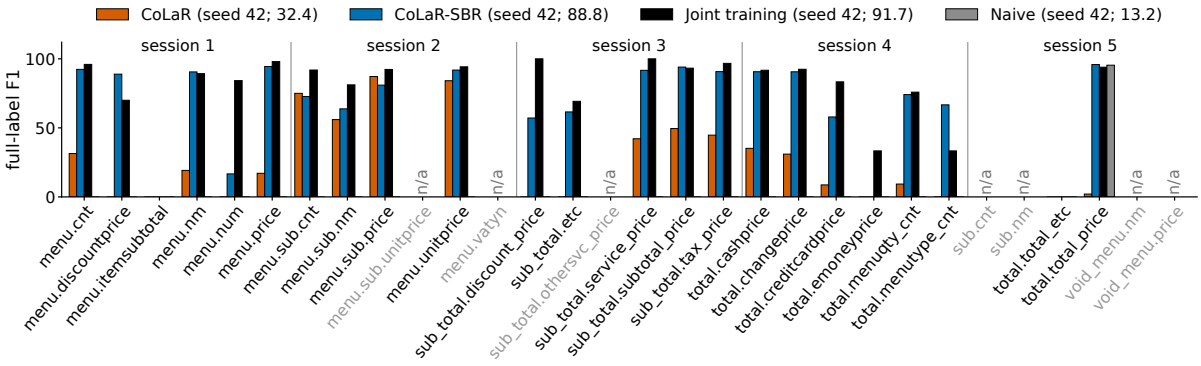


Figure 6. **Per-class full-label F1** of four prediction-dump re-runs (CIL-CORD, final model, unmasked test set; seed 42 only, each reproducing the full-label F1 of its clean run, given in the legend). Classes grouped by the session that introduces them; grey: the class does not occur in the test set.

least one session ends at zero; the trunk is frozen, so this is a readout effect, not drift. The 50-document memory supplies the positive counterweight: no session ends below the value in the last column, and isolation by deltas does not replace the memory across schemas (one seed). *Head slots.* LexSlot adds head slots that a session updates only when its OCR vocabulary selects them, and stores token ids, unlike CoLaR. Stacked on CoLaR-SBR it raises the gate mean and narrows the spread between seeds; we keep it as a row, not a method.

E. Drift evidence

Where drift sits. Under naive fine-tuning on a fixed label space every encoder forgets its earlier domains (Tab. 9). On our pilot stream FUNSD→CORD→SROIE, where each dataset brings its own labels, the representation drift grows with depth on all four encoders, largest at the head and the last layer and smallest at the embeddings (Fig. 8). That forgetting concentrates in the output-facing computation is known for image classifiers [31]; here it holds, on that stream, for three multimodal document encoders and a text-only one. On images the readout can also be repaired from class statistics (Appendix F); on token-labelled documents, in our experiments, only old-task supervision of the readout repaired it.

Replay against the classical remedies. On every encoder of Tab. 9 replay (ER, DER++) comes close to joint training

Table 7. **Component ablations on CIL-CORD**, the runs of Tab. 1 (clean protocol, LayoutLMv3, mean ± s.d. over three seeds). Ticks: latent memory (Mem.), blocked readout (Blk.), per-session deltas (Δ), true labels (True). All rows but naive, joint and the raw memory freeze at least the embeddings and layers 1..4 after session 1. MB: bytes kept across sessions (Sec. 5.4); SBR-Recall's deltas at two bytes per parameter, a reporting choice (113.2 as trained in fp32). ^{r}Stored documents given their true labels (H1): the older ones at each boundary, or all of them before each session (pre-session).

Configuration	Mem.	Blk.	Δ	True	Full-label F1	AA	MB
Naive					13.0±0.2	18.3±0.5	0
+ memory (CoLaR)	✓				30.7±1.5	33.5±1.0	100.7
+ blocked readout		✓			11.4±2.3	21.6±2.2	0
+ both (CoLaR-SBR)	✓	✓			89.1±2.0	88.4±2.6	100.7
+ both, raw memory	✓	✓			88.4±1.0	88.4±1.1	157.7
SBR-Recall		✓	✓		92.6±1.0	91.9±0.6	56.6
Relabel, boundary^{r}	✓			✓	76.5±2.3	63.6±1.9	100.9
Relabel, pre-session^{r}	✓			✓	92.1±1.3	91.3±1.3	100.9
Joint (reference)				✓	92.0±0.4	91.3±0.8	–

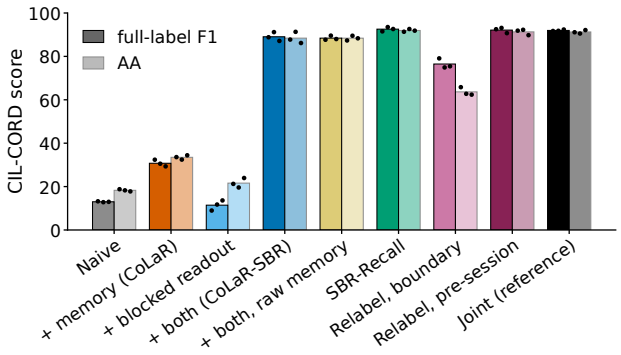


Figure 7. **The component grid of Tab. 7** (CIL-CORD, clean protocol; bar: mean of three seeds, dots: the seeds; full-label F1 dark, AA light).

Table 8. **Design-choice ablations** (fp32 gate runs†: 10-epoch cap, test-split early stopping, session-restricted scoring; AA per seed when fewer than three, else mean ± s.d.). The last two columns give the lowest diagonal entry (a session when learned) and the lowest final-row entry over every session and seed of the row. Generated.

Variant	AA	min $R[j,j]$	min $R[T,j]$
SBR-Recall delta capacity, CIL-CORD			
rank 8, Q/K/V, with memory	80.5†	61.5	63.2
rank 8, Q/K/V, no memory	82.5†	64.9	66.4
rank 64, Q/K/V + dense, no memory	92.8±0.5†	88.8	88.6
SBR-Recall across schemas, Mixed (rank 64)			
no memory	44.7†	82.0	0.0
50-document memory	85.9†	81.2	75.0
Head slots on CoLaR-SBR, CIL-CORD			
CoLaR-SBR	88.0±2.0†	85.9	72.8
+ LexSlot	89.7±0.5†	89.8	80.6

Table 9. **Drift and its remedy on four encoders** (DIL, AA, mean ± s.d. over three seeds; first grid, legacy protocol†). On average, replay comes within one point of joint training or exceeds it, and EWC and LwF stay within ten points of naive training. "–": did not fit the 6 GB GPU. Generated.

Backbone	Naive	EWC	LwF	ER	DER++	Joint
LayoutLMv3	41.3±1.5	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9	88.7±1.1
LiLT	40.1±0.5	48.4±5.9	–	85.7±0.5	84.4±0.3	83.8±1.2
BROS	40.3±1.8	45.0±4.9	35.1±3.1	87.8±0.4	87.5±1.1	87.5±1.3
BERT	39.4±0.3	46.6±3.0	40.5±0.1	77.1±0.9	77.4±0.6	77.2±0.4

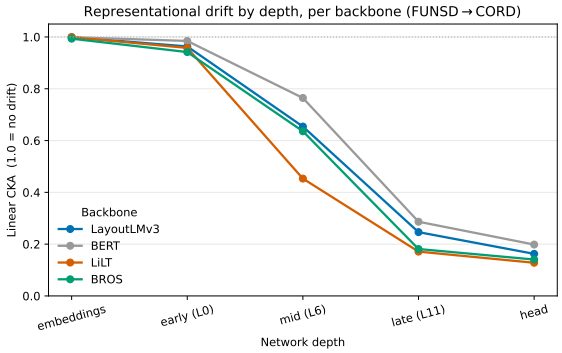


Figure 8. **Drift by depth**: linear CKA between checkpoints before and after the second task (FUNSD→CORD, each dataset with its own labels) under naive fine-tuning on four encoders (1.0: no drift; three seeds).

on average, whereas EWC and LwF stay near naive training; the benefit of EWC over naive training differs between encoders while the cause does not, which is why the table reports four. The buffer-free alternatives we built or ported for this drift each failed for a reason identified per boundary, in mostly single-seed diagnostics at their own operating points: head merging (TIES or Fisher) is close to a no-op; merging with Gaussian classifier realignment (LCA, three seeds) halves backward transfer but stops acquisition; subspace transfer (HGT, CUBER) trades stability for plasticity at the naive level; head slots without a buffer cannot re-ground the shared head; feature-Gaussian replay recovers only the densely labelled task, because a marginal summary breaks whole-document consistency; and refreshing stored labels with the model's own predictions is a no-op, because the model is confident on almost no stale "O" token.

F. Image classification

What transfers, and what does not. SLCA [44] and RanPAC [25] lead on both datasets, above the latent memory and CoLaR on every seed, and we make no state-of-the-art claim on images; our recipe is not SLCA's published one, and our SLCA and joint-training numbers are below the published ones. The latent memory is competitive with raw replay at 2.5× the items: on CIFAR-100 the latent memory and

Table 10. **Image class-incremental learning** (AA, mean ± s.d., seeds 42/7/123). CIFAR-100 and ImageNet-R in ten sessions (class order seed 1993, as in SLCA and RanPAC); ViT-B/16 pre-trained on ImageNet-21k with a linear head on the CLS token. House recipe, not the clean document protocol: AdamW at 5×10^{-5} (SLCA and LCA: SGD at 0.01), batch size 16, epoch cap 100 (SLCA: 20) with early stopping, fp32 with gradient checkpointing (the latent memory and CoLaR: mixed precision). ER and DER++ keep 2,000 raw images, SLCA class Gaussians, the latent memory and CoLaR 500 images per session as layer-4 latents (5,000 after ten sessions). SLCA and RanPAC take the slots CPFD and IS3 take on documents. ^{v}Diagnostic probe on seed-42 variant runs (on CIFAR-100 a second run of the latent memory, on ImageNet-R a CoLaR run with a 20-epoch cap): a ridge readout refit on all training data through the run's adapted trunk; not a method. Generated.

Method	CIFAR-100	ImageNet-R
Naive	11.7 ± 1.1	9.3 ± 0.3
LwF	15.7 ± 0.6	12.7 ± 1.4
ER (2,000 exemplars)	76.2 ± 1.2	54.8 ± 0.7
DER++ (2,000 exemplars)	76.5 ± 1.5	52.8 ± 1.1
CODA-Prompt	15.2 ± 1.8	8.4 ± 0.7
CL-LoRA (2025)	14.1 ± 1.2	11.4 ± 0.8
SLCA (2023)	87.5 ± 0.2	65.0 ± 0.3
RanPAC (2023)	85.3 ± 0.1	58.5 ± 0.3
LCA (2026)	77.2 ± 3.2	54.7 ± 3.6
Latent memory, 500/session (ours, uncompressed)	80.0 ± 0.7	55.1 ± 1.3
CoLaR r=128, 500/session (ours)	79.5 ± 1.0	55.3 ± 1.4
Ridge readout on a seed-42 variant run^{v} (probe)	88.3	72.3
Joint training (reference)	89.5 ± 0.4	77.6 ± 0.5

CoLaR are above ER and DER++ with 2,000 raw images on every seed, and on ImageNet-R they are within 2.7 and 1.6 points of ER on every seed. They store no pixel but more items (1,513 and 1,235 MB as recorded, against 1,204 MB for 2,000 raw images as our buffer stores them), and replay with 200 raw images falls to 42.3 / 21.6 (ER). Compressing each stored latent to rank 128 moves AA by at most 2.8 points on any seed. SLCA's margin comes from its classifier alignment: without it the same recipe falls to 38.0 / 25.8. On the one-seed variant runs of the probe row, the adapted trunk supports a readout above every seed of the latent memory and CoLaR rows, so the trunk holds more than the trained head reads out, the head-locus reading of documents; this rests on one seed. An image carries one label and has no background class, so there is no background conflict to remove: SBR is not run on images, and a stored image's label never goes stale. With a plain softmax head CoLaR reaches 79.5 / 55.3 against 89.5 / 77.6 for joint training, whereas on CIL-CORD the same memory with a shared softmax reaches 33.5 AA against 91.3, a much larger gap, consistent with a cause that only token labelling has.